

November

SPECIAL EVENTS

PHASES OF THE MOON

	Full Moon	New Moon
2007	November 24	November 9
2008	November 13	November 27
2009	November 2	November 16
2010	November 21	November 6
2011	November 10	November 25
2012	November 28	November 13

ECLIPSES

2011: November 25 A partial eclipse of the Sun is visible from the southern Indian Ocean and Antarctica.

2012: November 13–14 A total eclipse of the Sun is visible from northeastern Australia and the South Pacific. A partial eclipse is visible from New Zealand, the rest of Australia, and the Pacific Ocean.

THE PLANETS

2007: November 8
Mercury is at greatest morning elongation, magnitude -0.4 .

2008: November 30
Venus and Jupiter appear four Moon-widths apart in the western evening sky.

2010: November 20
Mercury and Mars appear about three Moon-widths apart in the western evening sky.

2011: November 1–14
Mercury and Venus appear four Moon-widths apart in the western evening sky.

2011: November 14
Mercury is at greatest evening elongation, magnitude -0.2 .

2012: November 27
Venus and Saturn appear one Moon-width apart in the eastern dawn sky.

Elongation is explained on *page 14*

Two celebrated stars—Mira in Cetus, the sea monster, and Algol in Perseus—are in the sky for both northern and southern observers. Cetus and Perseus are joined by the constellation of Andromeda, the princess rescued from Cetus by Perseus. Northern-latitude sky watchers see the Milky Way arching overhead, and southern-latitude observers see four birds as they await the arrival of the summer stars.

NORTHERN LATITUDES

Perseus and Andromeda are high overhead. Andromeda's parents, Cepheus and Cassiopeia, are to the north. Andromeda is well placed for observing the Andromeda Galaxy. Algol in Perseus is an eclipsing binary—two stars that exist together and orbit each other. As the dimmer of the pair passes in front of the brighter, the combined brightness drops. Algol's magnitude changes from 2.1 to 3.4 and back again.

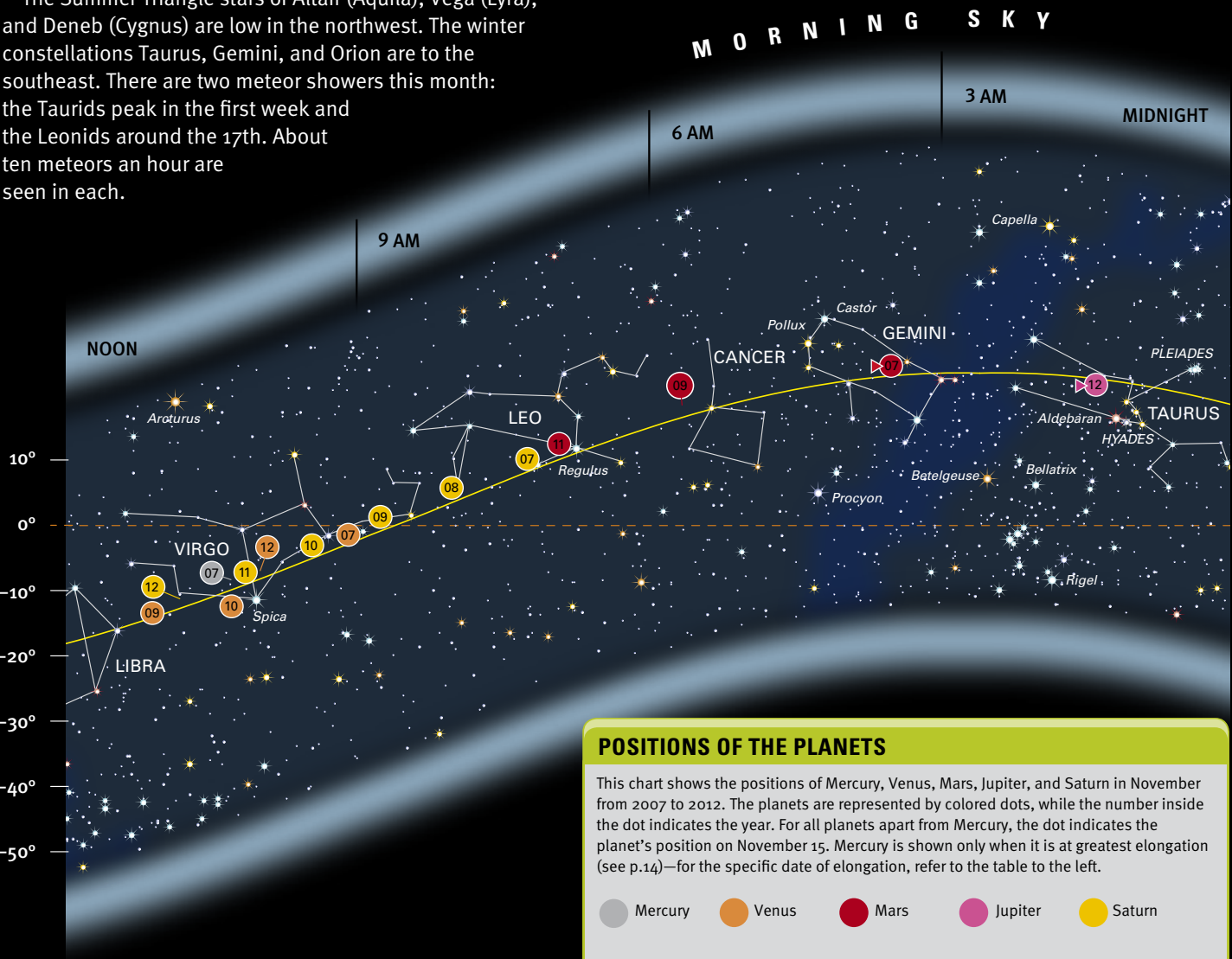
Cetus is due south. Its red giant Mira is a variable star whose brightness varies in a cycle lasting 11 months. It changes from a naked-eye star of magnitude 3, to a star of magnitude 10 visible only through a telescope. Pisces is above Cetus; to its right is the Great Square of Pegasus.

The Summer Triangle stars of Altair (Aquila), Vega (Lyra), and Deneb (Cygnus) are low in the northwest. The winter constellations Taurus, Gemini, and Orion are to the southeast. There are two meteor showers this month: the Taurids peak in the first week and the Leonids around the 17th. About ten meteors an hour are seen in each.



Leonid meteor shower

Thirty images are combined in this view of a Leonid meteor shower. Meteors are best seen about 50 degrees above the horizon and 30–40 degrees to one side of the point from which they appear to radiate—in this case, in Leo.



POSITIONS OF THE PLANETS

This chart shows the positions of Mercury, Venus, Mars, Jupiter, and Saturn in November from 2007 to 2012. The planets are represented by colored dots, while the number inside the dot indicates the year. For all planets apart from Mercury, the dot indicates the planet's position on November 15. Mercury is shown only when it is at greatest elongation (see p.14)—for the specific date of elongation, refer to the table to the left.

Mercury Venus Mars Jupiter Saturn

EXAMPLES  Mars's position on November 15, 2009  Saturn's position on November 15, 2009. The arrow indicates that the planet is in retrograde motion (see p.14)